

Executive Summary

The problem. A security model can score better while letting more attacks pass without a warning. Calling an attack the wrong type can still prompt an investigation. Calling it benign can hide it. A single overall score does not show that difference.

What the main experiment found. In one comparison, the overall score improved, but the share of exfiltration records receiving any attack warning fell from about 85% to 76%. The size of the change varied across training runs. The same policy ordering did not repeat on the AIT test source.

What helped. A simple OR rule keeps a warning whenever any member model flags an attack. It stops the combination step from erasing an available warning. It cannot help when every member calls the event benign.

What the repair tests found. Adding a TCP/22 policy rule raised UNRAVELED episode coverage from 6 of 18 to 18 of 18. It also added 510 benign-labeled flow warnings and 460 grouped investigation cases. The twelve recovered episodes were small, time-defined groups from one endpoint pair, not twelve confirmed theft incidents. On AIT, exfiltration used UDP/53, so the unchanged TCP/22 rule added no exfiltration coverage.

What to do with this result. Review model updates using attack warnings, stage accuracy and benign workload together. Keep mandatory policy decisions explicit so a model cannot override them. Test the policy on traffic it covers and on other attack channels. The study supports this review method and a limited local repair; it does not establish a universal exfiltration detector or prove that theft was prevented.